Career Research to Portfolio Design

IS373 | Independent student guide | November 3, 2026

Career research helps you design your individual hosted portfolio. Explore the work behind AI-related titles, choose a provisional direction, and use employer evidence to decide your audience, content, design and learning priorities. Your RAG application and group contribution supply project evidence for that portfolio.

1. Discover options before choosing

Start with FDE and explore at least two alternatives. Search beyond frontier AI labs: established software firms, finance, consulting, health organizations and other industries may use different titles for related work.

Role family	Search phrases	Work to investigate
Forward deployed engineering	FDE, forward deployed software engineer, AI implementation	Customer discovery, integration, delivery and adoption
AI engineering	Applied AI engineer, AI software engineer, LLM engineer	Applications, retrieval, evaluation and reliable integration
AI product engineering	AI product engineer, product engineer AI	User needs, full-stack features, feedback and iteration
Platform and operations	AI platform, ML platform, MLOps, inference, AgentOps	Deployment, observability, developer tools and reliability
AI security	Agent security, AI application security, ML security	Threat models, access boundaries, controls and security testing
Data, search and evaluation	Retrieval engineer, search, data engineer AI, AI evaluation	Content quality, retrieval relevance, datasets and evaluation

These are search families, not standardized titles or proof of job growth. A product engineer may build software; a product manager may define priorities. Read the responsibilities before grouping roles together.

First research session

Choose three families that interest you, search for original employer descriptions and save candidate links. Do not narrow your choice based only on salary, company fame or an AI-generated description. Check level and eligibility early.

Copyable prompt: discover roles

Help me explore FDE, AI engineering, AI product engineering, platform/MLOps, AI security and related roles. My interests are [interests] and my current evidence is [projects/skills]. Suggest three families to investigate, useful search phrases and questions about daily work. Distinguish hypotheses about fit from facts in employer descriptions. Do not invent jobs or claim a title is growing without evidence.

2. Collect original employer evidence

For the Plan review, read three original employer postings spanning at least two role families. Include FDE as a reference and at least one alternative. Try to include a verified internship/junior opportunity. A senior example can inform a learning plan, but label it as a longer-term reference. Recheck availability when applying.

Keep a private research record. For each posting save:

- Employer, exact title, employer industry and original URL.
- Date checked; posted date/deadline if visible; current status or not confirmed.
- Location, work arrangement, travel, level and eligibility.
- Three responsibilities in your own words.
- Required versus preferred qualifications.
- One short excerpt and its section supporting your interpretation.
- One relevant thing you have actually done, one gap and one small next action.
- A readable screenshot or saved evidence note and any uncertainty.

Copyable prompt: inspect a posting

@Chrome Read [original employer URL]. Record employer, exact title, date checked, location, level, eligibility, responsibilities and required versus preferred skills. Save a readable screenshot of the relevant sections. Separate what the page says from your interpretation. Mark missing information not stated. Do not apply, create an account, contact anyone or invent my experience.

Check the evidence yourself

Open every original page and confirm that it supports the notes. Search results can be stale, salary ranges depend on context, and years of experience may rule out a near-term application. Keep the title as written even if it does not contain AI. Three postings describe your sample, not the whole labor market.

Manual fallback

If browser/computer tools are unavailable, browse manually, save screenshots and attach them for analysis. Record URLs and dates yourself. You do not need a paid tool subscription to follow this research process.

3. Compare directions and choose a reader

Compare two possible directions using your employer records. Identify what interests you about the actual work, what you can already demonstrate, what remains missing and how you will investigate further. A provisional choice can change as you learn.

Comparison question	Your evidence
What would I do regularly?	Tasks from the original descriptions
What attracts me or feels unsuitable?	Your own reasoning, including travel/work format
What can I demonstrate today?	Actual project links, implementation and checks
Which gap can this course help address?	A bounded feature, evaluation or delivery task
What is my next step and when?	A dated action small enough to complete

Write a short private reflection, approximately 150-250 words or equivalent concise bullets. Explain the two directions, your leaning, evidence, one gap, a dated action and something the research changed about your assumptions. Link or summarize its relevant design decisions in the individual project record.

Translate responsibilities into portfolio evidence

Responsibility	Evidence you could develop in IS373
Understand a user's problem	Reader/task statement, discovery questions, rejected scope and observable story
Build with retrieval	Source manifest, chunks or graph model, query trace and a justified architecture
Evaluate an AI application	Expected evidence, actual outputs, missing-evidence case and an improvement/retest
Operate reliable services	Hosted app, persistent data, reproducible setup and an observed recovery check
Protect data and access	Server-side import permissions, private credentials and a relevant denied-request check
Deliver with a team	Named contribution, issue/PR, cross-review and integrated group result

These are learning connections, not proof of professional production experience. Keep claims proportional to what you actually built and checked.

4. Design the portfolio from your findings

Choose a primary reader: for example, an internship reviewer hiring for AI application development. Write one sentence describing what that person should understand after viewing your site. Map three responsibilities from your postings to concrete evidence you can feature.

Plan a small content structure:

- Introduction: interests, provisional direction and accurate capabilities.
- Individual RAG case study and working chat.
- Group case study that names your own contribution.
- Repository/demo/check links and a clear explanation of decisions and limits.
- Suitable resume, GitHub/LinkedIn links and a professional contact route.

Compare two lightweight layouts or case-study treatments using the same material. Choose one with a reason. You do not need two complete portfolios. Your target role should affect what you emphasize: an application role may prioritize user tasks and answer quality; platform interests may emphasize deployment and observability; security interests may emphasize access boundaries and evaluated risks.

Use design lenses as hypotheses

Choose a clear visual hierarchy, readable typography and a suitable voice. A brand archetype can help describe your intended professional character. Cialdini can help you consider honest persuasion: demonstrate expertise with inspectable work, or show attributed feedback only when permission allows. Never invent endorsements, credentials or scarcity. Modernist clarity or a more expressive postmodern treatment should serve the reader and accessibility.

Ask a real peer to try it

Give a peer three tasks: identify your intended role, find evidence of one capability and try the RAG chat. Observe without coaching. Record a real issue, make a meaningful change and retest. ChatGPT can critique a draft, but AI roleplay does not count as human peer review.

Copyable prompt: derive the portfolio plan

Using my verified job records and actual project evidence, propose a target reader, three responsibility-to-evidence mappings, a site outline and two lightweight design options. Explain each recommendation from the evidence. Separate public portfolio content from private reflection. Do not invent expertise, testimonials, experience or finished features.

5. Bring the research into the review cycle

Nov 3/5: bring one posting and initial interests. Nov 10/12: compare roles and discuss portfolio direction. Nov 17 Plan: include the three posting records, comparison, target reader, evidence map and portfolio outline in the combined individual record. Continue Build Nov 24, checked hosted release Dec 3 and Improve Dec 10. Canvas supplies exact instructions and submissions.

Career research is the portfolio's research stage, not another final deliverable. Keep one individual record with portfolio/career and RAG sections. Publish suitable goals and project evidence while keeping personal reflections private.

Source leads

The following original employer examples were checked November 3, 2026. They are starting points, not a guarantee of availability or student eligibility. Read the entire current description before using it.

- OpenAI: Forward Deployed Engineer: discovery, scoping, building and production rollout. This reference requires substantial experience and travel.
- Citi: Senior AI Engineer, Banking Technology: a senior AI engineering example outside an AI lab, in Jersey City.
- Anthropic: Product Engineer, Computer Use: end-to-end product delivery across interface, agent runtime and services.
- OpenAI: Software Engineer, Compute Infrastructure: reliability, systems and infrastructure tooling.
- OpenAI: Security Engineer, Agent Security: threat modeling, isolation, controls and monitoring.
- NJIT Career Development Services: additional opportunities and advising.

Final self-check

Can you open all three employer sources? Can you explain the tasks and eligibility? Does your portfolio audience follow from the research? Does each claim point to work you actually did? Do you have one gap and a dated action? Can a reader distinguish your own contribution from the team's work?

Portfolio brief | Reviews and rubric | Project record